ROOTS BLOWER

RL SERIES

Three-lobe roots blower

RANGE CHARACTERISTICS

Bore	**40 – 300 mm (1½" – 12")**
Capacity	**0.1 – 178 m³/min**
Pressure	**1000 – 8000 mmAq (9.8 – 78.5 kPa)**
Max. pressure	**10,000 mmAq (single stage)**
Speed	**600 – 1900 r/min**
Required power	**0.25 – 250.5 kW**
Noise	**56 – 111 dB(A) at 1 m**
Drive	**V-belt (standard), direct coupling**

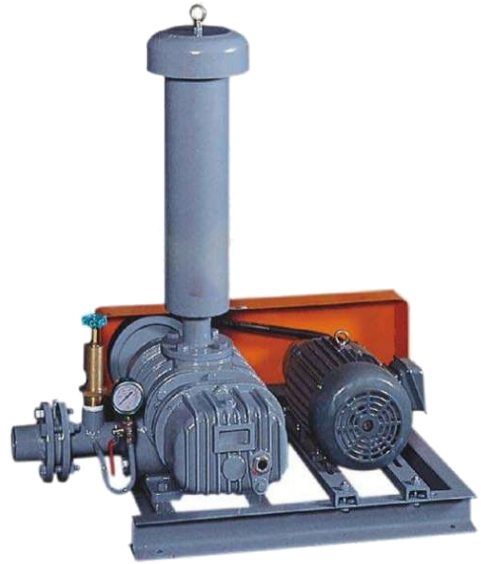
MODEL CODE

RL-…	Blower body only
RLC-…	Complete set (pressure)
RLV-…	Complete set (vacuum)

THREE-LOBE ROOTS BLOWER

Pressure and vacuum operation | bore 40 – 300 mm





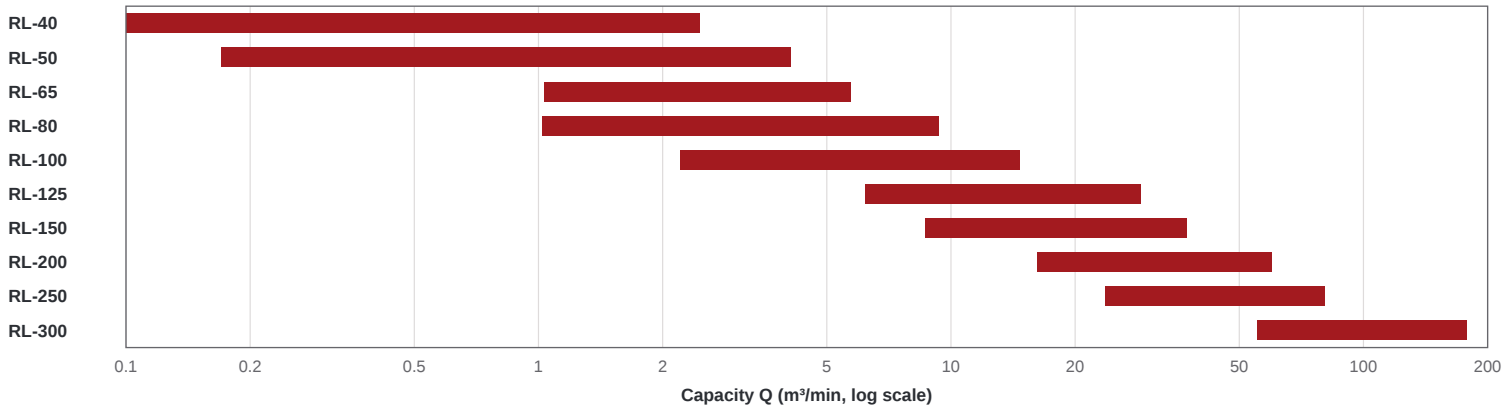
FEATURES

- Three-lobe rotors: low pressure pulsation, low noise and vibration.
- Oil-free air: no lubricant in the compression chamber.
- Oil-lubricated bearings and timing gears.
- Cast-iron casing FC25; rotors with SCM440 shafts; SNCM220 timing gears.
- Rotors machined on precision NC machines and dynamically balanced.
- Complete sets with silencer, safety valve, check valve and gauge.

APPLICATIONS

- Aeration in wastewater treatment
- Aquaculture and fish ponds
- Agitation of liquids
- Filter backwash
- Pneumatic conveying
- Combustion air
- Drying and air blowing
- Vacuum duties (RLV complete sets)

CAPACITY RANGE BY MODEL



Construction

Three-lobe roots blower

ADVANTAGES

With the three-lobe rotor, the reverse-flow pressure variation period is only 2/3 of that of a conventional two-lobe rotor and the peak pressure is also lower, so noise and pressure pulsation are greatly reduced.

1. Less back flow, steady air flow, lower vibration and noise.
2. Lower discharge pressure pulsation means lighter loads on bearings and timing gears and a longer service life.
3. At the same speed, the three-lobe blower delivers a larger air flow than a conventional two-lobe blower.
4. Adequate clearance between the rotors, and between the rotors and the casing, ensures no contact during operation and higher efficiency.
5. Rotor precision is fully controlled and the variation between blowers is almost nil, because the rotors are machined on precision NC machines.
6. The rotors are dynamically balanced during manufacture, so they run almost free of vibration.

Redpump Italy Standard

GENERAL

Redpump three-lobe roots blowers, also known as positive displacement rotary lobe blowers, are used in conditions ranging from vacuum to high pressure in all branches of industry, especially in wastewater treatment plants. The simple design, easy handling and stable performance make a wide range of applications possible. Their main characteristic is to deliver a constant flow of air at different pressures, and efficiency is normally high.

Another important feature is that the conveyed air is not contaminated, because there is no lubricant in the compression chamber.

The blower consists of a cast-iron casing in which two rotors of conjugate shape rotate, synchronized by a set of timing gears. During rotation there is no contact between the lobes, or between the lobes and the casing. Front and rear side covers close the casing; the shaft bearings and the seals that prevent air leakage are located there.

OPERATING PRINCIPLE

The two three-lobe rotors rotate in opposite directions, synchronized by a set of timing gears. Thanks to the geometry of the lobes the rotors always share a common generatrix, which, with very small machine clearances, acts as a seal between them. During operation one or two lobes of each rotor form a chamber with the inner surface of the casing in which the air is trapped. As rotation continues, the trapped air moves on until it reaches the discharge port. The two rotors make six intake and discharge cycles per revolution, so the capacity of the blower is determined by its speed and rises in proportion to it, while it is largely independent of the pressure difference between inlet and outlet.

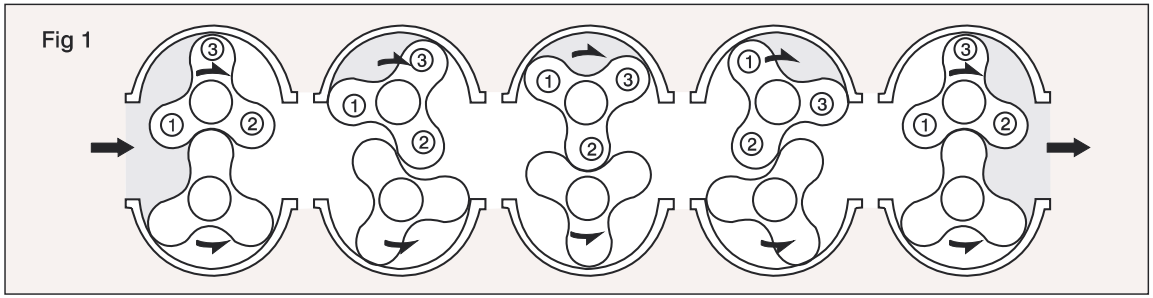
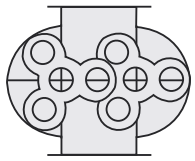


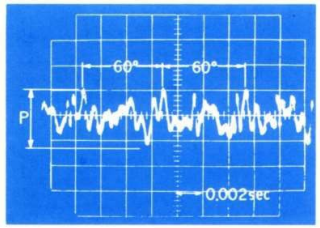
Fig. 1 — one revolution: intake (left), compression and discharge (right)

3-Lobes Cylinder

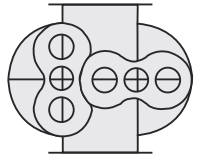


(Fig.1)

Three-lobe rotor — discharge pressure

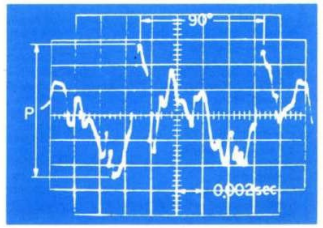


2-Lobes Cylinder



(Fig.2)

Two-lobe rotor — discharge pressure



Discharge pressure at 5000 mmAq and 1560 r/min for both blowers.

Characteristics

Max. performance, model range and complete set

MODEL CODE

RL: blower body only.

RLC: complete set — blower with standard accessories for pressure operation.

RLV: complete set — blower with standard accessories for vacuum operation.

LUBRICATION

Front bearing	**Oil**
Rear bearing	**Oil**
Gear box	**Oil**

PRESSURE RANGE

Recommended operation: 1000 – 8000 mmAq (9.8 – 78.5 kPa). 1 atm = 10333 mmAq = 760 mmHg.

Redpump Italy Standard

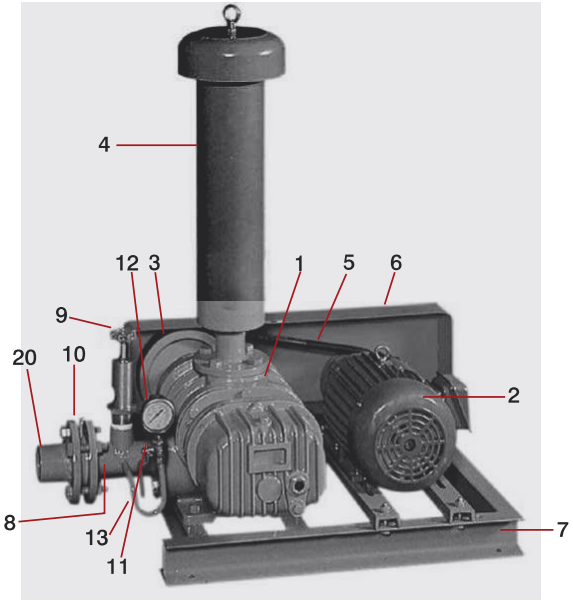
MAXIMUM PERFORMANCE

Type	Max. pressure	Max. vacuum
RL single stage	10,000 mmAq (1.0 kg/cm²)	-6,700 mmAq (-500 mmHg)
RL double stage	15,000 mmAq (1.5 kg/cm²)	-8,100 mmAq (-600 mmHg)

MODEL RANGE

Bore (mm)	Blower body	Complete set pressure	Complete set vacuum
40	RL-40	RLC-40	–
50	RL-50	RLC-50	RLV-50
65	RL-65	RLC-65	RLV-65
80	RL-80	RLC-80	RLV-80
100	RL-100	RLC-100	RLV-100
125	RL-125	RLC-125	RLV-125
150	RL-150	RLC-150	RLV-150
200	RL-200	RLC-200	RLV-200
250	RL-250	RLC-250	RLV-250
300	RL-300	RLC-300	RLV-300

COMPLETE SET (RLC)



No.	Standard accessories
1	Blower Body
2	Motor
3	Blower Pulley
4	Suction Silencer Set (with air filter & cover)
5	V-belt
6	Belt Cover
7	Base Plate
8	T-type Reducer
9	Pressure Safety Valve
10	Check Valve
11	Ball Valve
12	Pressure Gauge
13	Gauge Pipe
20	Discharge Flange (optional)

RL SERIES

RL-40

Three-lobe roots blower



KEY DATA

Bore	**1½" (40 mm)**
Speed	**800 – 1900 r/min**
Capacity	**0.1 – 2.45 m³/min**
Pressure	**1000 – 8000 mmAq (9.8 – 78.5 kPa)**
Required power	**0.25 – 4.26 kW**
Noise	**56 – 75 dB(A) at 1 m**
Weight	**65 kg body / 103 kg complete set**

MODEL CODE

RL-40
Blower body only

RLC-40
Complete set, pressure operation

PERFORMANCE DATA

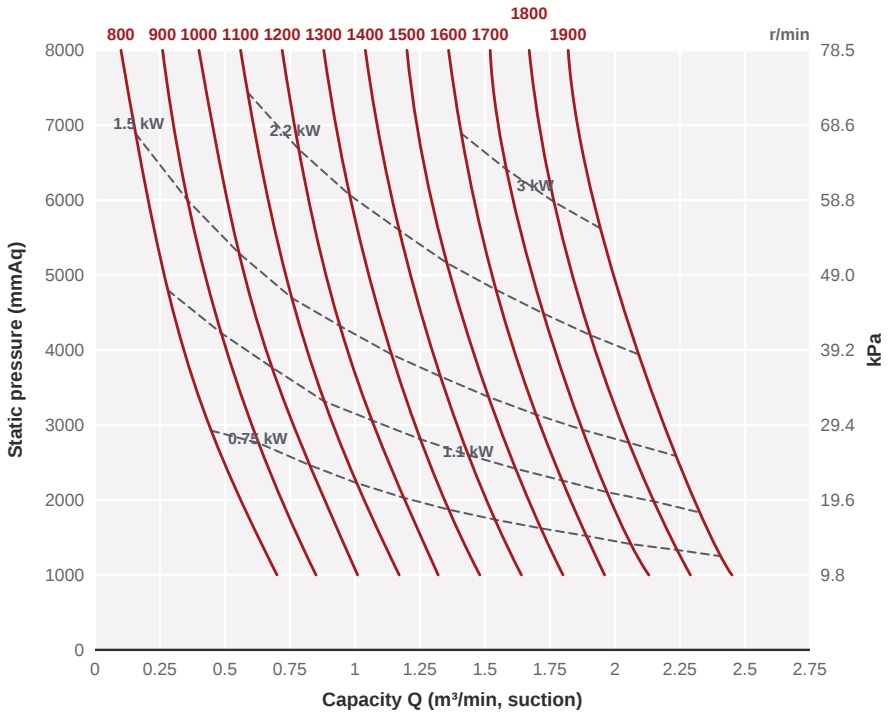
Bore 1½" (40 mm) | 800 – 1900 r/min | pressure operation

PERFORMANCE TABLE

	1000 mmAq 9.8 kPa		2000 mmAq 19.6 kPa		3000 mmAq 29.4 kPa		4000 mmAq 39.2 kPa		5000 mmAq 49.0 kPa		6000 mmAq 58.8 kPa		7000 mmAq 68.6 kPa		8000 mmAq 78.5 kPa	
r/min	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW
800	0.70	0.25	0.56	0.50	0.45	0.77	0.35	0.94	0.26	1.14	0.21	1.34	0.15	1.52	0.10	1.70
900	0.85	0.28	0.71	0.56	0.61	0.81	0.50	1.05	0.42	1.27	0.36	1.50	0.30	1.71	0.26	1.91
1000	1.01	0.31	0.87	0.62	0.75	0.90	0.67	1.16	0.58	1.43	0.50	1.67	0.46	1.90	0.40	2.12
1100	1.17	0.35	1.03	0.68	0.92	0.99	0.82	1.34	0.73	1.57	0.67	1.84	0.61	2.09	0.56	2.34
1200	1.32	0.36	1.19	0.74	1.08	1.08	0.98	1.40	0.88	1.71	0.83	2.01	0.77	2.29	0.72	2.45
1300	1.48	0.41	1.34	0.80	1.23	1.17	1.14	1.52	1.05	1.86	0.98	2.18	0.93	2.47	0.88	2.56
1400	1.64	0.44	1.51	0.86	1.40	1.26	1.29	1.63	1.22	1.99	1.14	2.34	1.08	2.66	1.04	2.77
1500	1.80	0.47	1.66	0.92	1.56	1.35	1.45	1.75	1.36	2.14	1.30	2.51	1.24	2.85	1.20	2.98
1600	1.96	0.50	1.82	0.98	1.72	1.44	1.61	1.87	1.52	2.28	1.46	2.67	1.41	3.04	1.36	3.30
1700	2.13	0.54	1.98	1.05	1.87	1.53	1.77	1.99	1.68	2.42	1.61	2.84	1.55	3.23	1.52	3.67
1800	2.29	0.57	2.14	1.11	2.03	1.62	1.93	2.11	1.83	2.56	1.77	3.01	1.71	3.42	1.67	3.93
1900	2.45	0.60	2.30	1.20	2.19	1.71	2.09	2.23	1.99	2.70	1.92	3.18	1.86	3.61	1.82	4.26

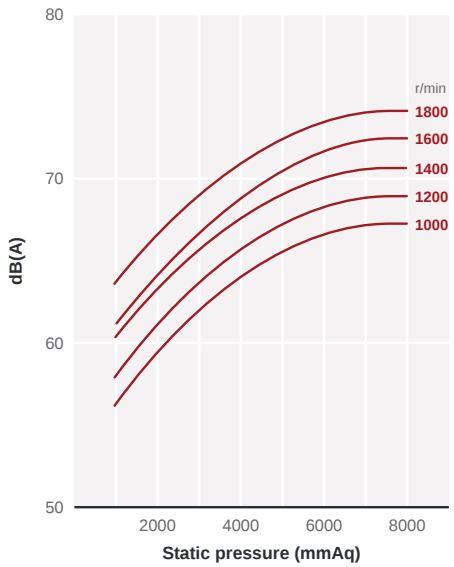
Q: air volume at suction (m³/min). kW: required power including V-belt transmission — no extra margin needed when selecting the motor.

PERFORMANCE CURVES



Dashed lines: motor power steps, interpolated from the power values of the table.

NOISE LEVEL · dB(A) at 1 m



RL SERIES

RL-50

Three-lobe roots blower



KEY DATA

Bore	**2" (50 mm)**
Speed	**800 – 1900 r/min**
Capacity	**0.17 – 4.08 m³/min**
Pressure	**1000 – 8000 mmAq (9.8 – 78.5 kPa)**
Required power	**0.42 – 7.09 kW**
Noise	**61 – 80 dB(A) at 1 m**
Weight	**70 kg body / 136 kg complete set**

MODEL CODE

RL-50
Blower body only

RLC-50
Complete set, pressure operation

RLV-50
Complete set, vacuum operation

PERFORMANCE DATA

Bore 2" (50 mm) | 800 – 1900 r/min | pressure operation

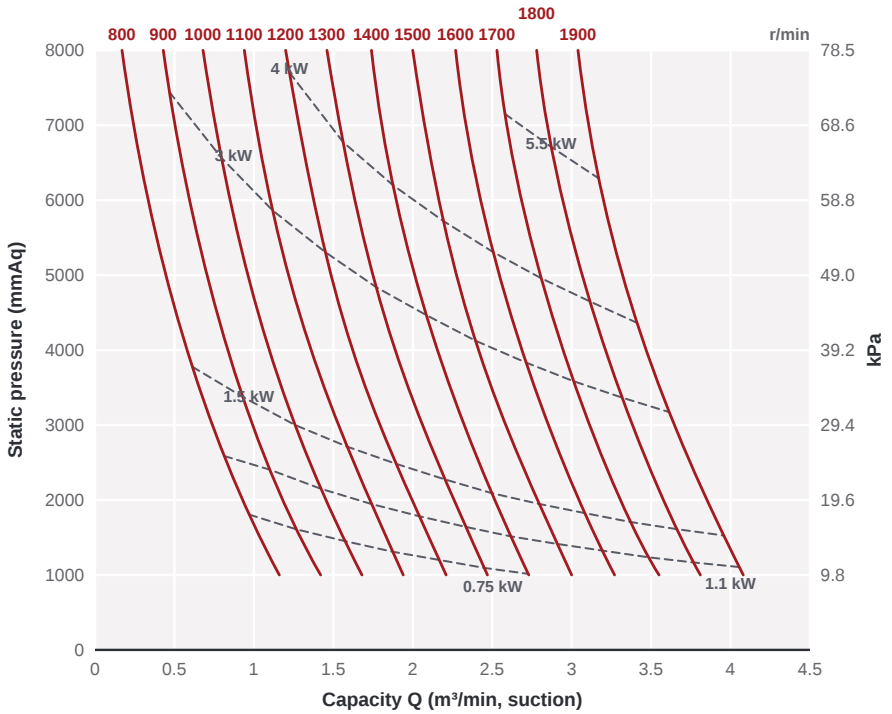
Redpump Italy Standard

PERFORMANCE TABLE

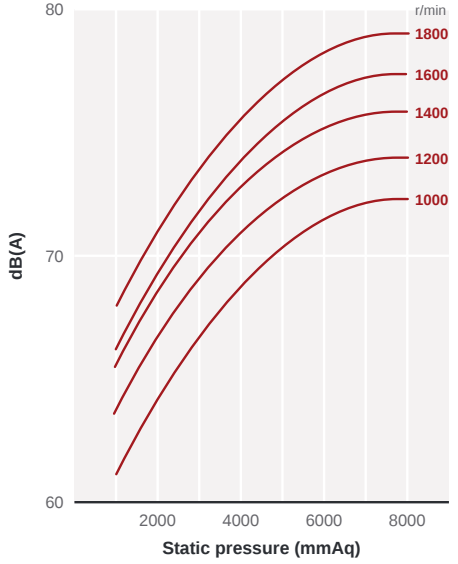
	1000 mmAq 9.8 kPa		2000 mmAq 19.6 kPa		3000 mmAq 29.4 kPa		4000 mmAq 39.2 kPa		5000 mmAq 49.0 kPa		6000 mmAq 58.8 kPa		7000 mmAq 68.6 kPa		8000 mmAq 78.5 kPa	
r/min	**Q**	**kW**	**Q**	**kW**	**Q**	**kW**	**Q**	**kW**	**Q**	**kW**	**Q**	**kW**	**Q**	**kW**	**Q**	**kW**
800	1.16	0.42	0.92	0.83	0.75	1.29	0.58	1.56	0.44	1.90	0.34	2.22	0.24	2.53	0.17	2.83
900	1.42	0.47	1.18	0.93	1.00	1.35	0.84	1.75	0.70	2.12	0.60	2.50	0.50	2.85	0.43	3.19
1000	1.68	0.52	1.45	1.03	1.26	1.50	1.10	1.94	0.96	2.38	0.85	2.78	0.76	3.17	0.68	3.54
1100	1.94	0.58	1.72	1.13	1.53	1.65	1.37	2.24	1.21	2.62	1.11	3.06	1.02	3.49	0.94	3.90
1200	2.21	0.63	1.99	1.23	1.80	1.79	1.63	2.33	1.48	2.85	1.37	3.34	1.29	3.81	1.20	4.08
1300	2.47	0.69	2.26	1.33	2.06	1.95	1.90	2.53	1.74	3.09	1.64	3.62	1.55	4.11	1.46	4.27
1400	2.73	0.74	2.51	1.44	2.33	2.10	2.15	2.72	2.01	3.32	1.90	3.89	1.81	4.44	1.74	4.62
1500	3.00	0.79	2.78	1.54	2.59	2.25	2.42	2.92	2.27	3.56	2.17	4.17	2.07	4.76	2.00	4.96
1600	3.27	0.84	3.04	1.64	2.86	2.40	2.69	3.12	2.54	3.80	2.43	4.45	2.34	5.08	2.27	5.46
1700	3.55	0.90	3.30	1.75	3.12	2.55	2.94	3.32	2.80	4.03	2.69	4.73	2.59	5.39	2.53	6.12
1800	3.81	0.95	3.57	1.85	3.39	2.70	3.21	3.52	3.06	4.26	2.95	5.01	2.85	5.70	2.78	6.56
1900	4.08	1.00	3.84	1.95	3.65	2.85	3.48	3.72	3.31	4.49	3.21	5.29	3.11	6.01	3.04	7.09

Q: air volume at suction (m³/min). kW: required power including V-belt transmission — no extra margin needed when selecting the motor.

PERFORMANCE CURVES



NOISE LEVEL · dB(A) at 1 m



Dashed lines: motor power steps, interpolated from the power values of the table.

RL SERIES

RL-65

Three-lobe roots blower



KEY DATA

Bore	**2½" (65 mm)**
Speed	**800 – 1900 r/min**
Capacity	**1.03 – 5.71 m³/min**
Pressure	**1000 – 8000 mmAq (9.8 – 78.5 kPa)**
Required power	**0.58 – 9.27 kW**
Noise	**61 – 80 dB(A) at 1 m**
Weight	**75 kg body / 155 kg complete set**

MODEL CODE

RL-65
Blower body only

RLC-65
Complete set, pressure operation

RLV-65
Complete set, vacuum operation

PERFORMANCE DATA

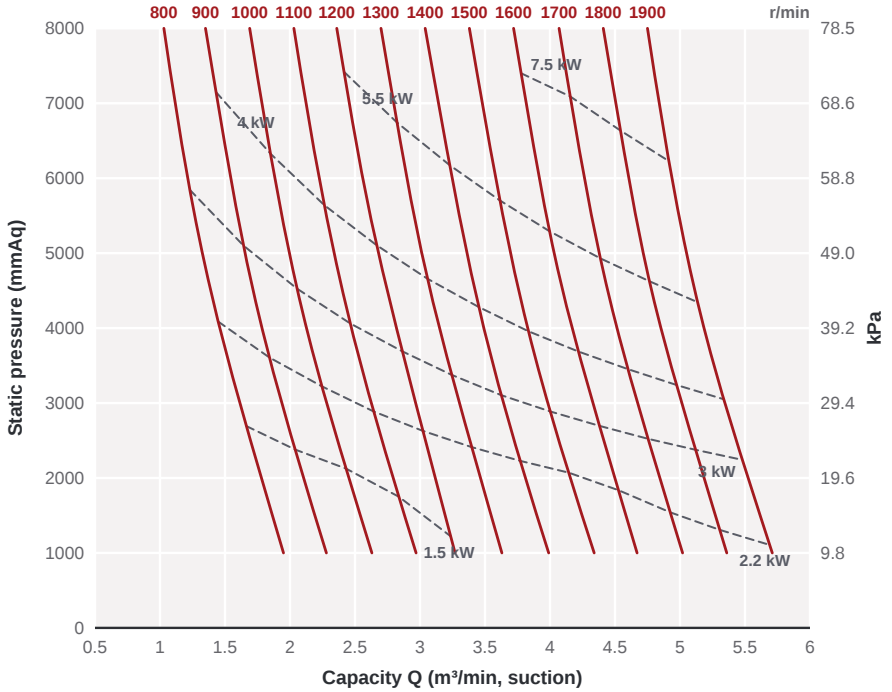
Bore 2½" (65 mm) | 800 – 1900 r/min | pressure operation

PERFORMANCE TABLE

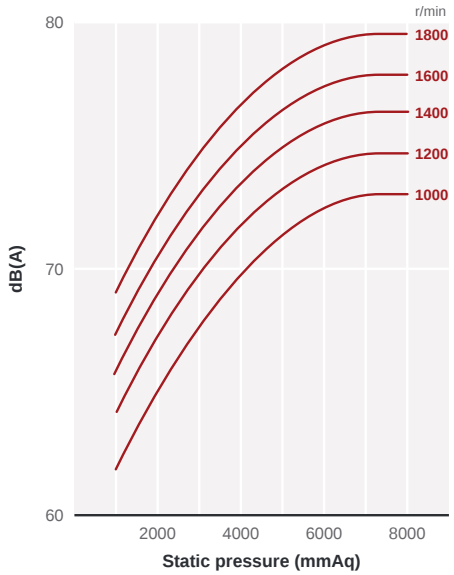
	1000 mmAq 9.8 kPa		2000 mmAq 19.6 kPa		3000 mmAq 29.4 kPa		4000 mmAq 39.2 kPa		5000 mmAq 49.0 kPa		6000 mmAq 58.8 kPa		7000 mmAq 68.6 kPa		8000 mmAq 78.5 kPa	
r/min	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW
800	1.95	0.58	1.78	1.14	1.61	1.66	1.46	2.16	1.33	2.62	1.21	3.07	1.12	3.49	1.03	3.91
900	2.28	0.79	2.11	1.28	1.94	1.86	1.79	2.42	1.65	2.95	1.54	3.45	1.45	3.93	1.35	4.40
1000	2.63	1.14	2.46	1.42	2.29	2.06	2.12	2.69	1.99	3.28	1.89	3.83	1.78	4.35	1.69	4.89
1100	2.97	1.29	2.80	1.57	2.62	2.28	2.47	2.96	2.34	3.61	2.23	4.22	2.12	4.81	2.03	5.38
1200	3.27	1.44	3.14	1.71	2.96	2.49	2.80	3.23	2.68	3.93	2.56	4.60	2.46	5.24	2.36	5.86
1300	3.63	1.54	3.48	1.85	3.29	2.70	3.14	3.50	3.01	4.26	2.90	4.98	2.80	5.68	2.70	6.35
1400	3.99	1.64	3.82	1.99	3.64	2.91	3.50	3.77	3.36	4.59	3.24	5.37	3.15	6.12	3.04	6.84
1500	4.34	1.74	4.16	2.13	3.98	3.11	3.83	4.04	3.70	4.91	3.58	5.75	3.48	6.55	3.38	7.32
1600	4.67	1.84	4.51	2.27	4.32	3.32	4.17	4.31	4.03	5.24	3.92	6.13	3.82	6.98	3.72	8.28
1700	5.02	1.94	4.86	2.42	4.66	3.53	4.52	4.57	4.37	5.57	4.26	6.51	4.17	7.42	4.07	8.30
1800	5.36	2.04	5.20	2.56	5.00	3.73	4.86	4.85	4.71	5.89	4.60	6.89	4.50	7.85	4.41	8.78
1900	5.71	2.14	5.53	2.70	5.34	3.94	5.19	5.12	5.05	6.22	4.94	7.27	4.84	8.28	4.75	9.27

Q: air volume at suction (m³/min). kW: required power including V-belt transmission — no extra margin needed when selecting the motor.

PERFORMANCE CURVES



NOISE LEVEL · dB(A) at 1 m



Dashed lines: motor power steps, interpolated from the power values of the table.

RL SERIES

RL-80

Three-lobe roots blower



KEY DATA

Bore	**3" (80 mm)**
Speed	**800 – 1900 r/min**
Capacity	**1.02 – 9.32 m³/min**
Pressure	**1000 – 8000 mmAq (9.8 – 78.5 kPa)**
Required power	**1.4 – 14.14 kW**
Noise	**62 – 82 dB(A) at 1 m**
Weight	**143 kg body / 250 kg complete set**

MODEL CODE

RL-80
Blower body only

RLC-80
Complete set, pressure operation

RLV-80
Complete set, vacuum operation

PERFORMANCE DATA

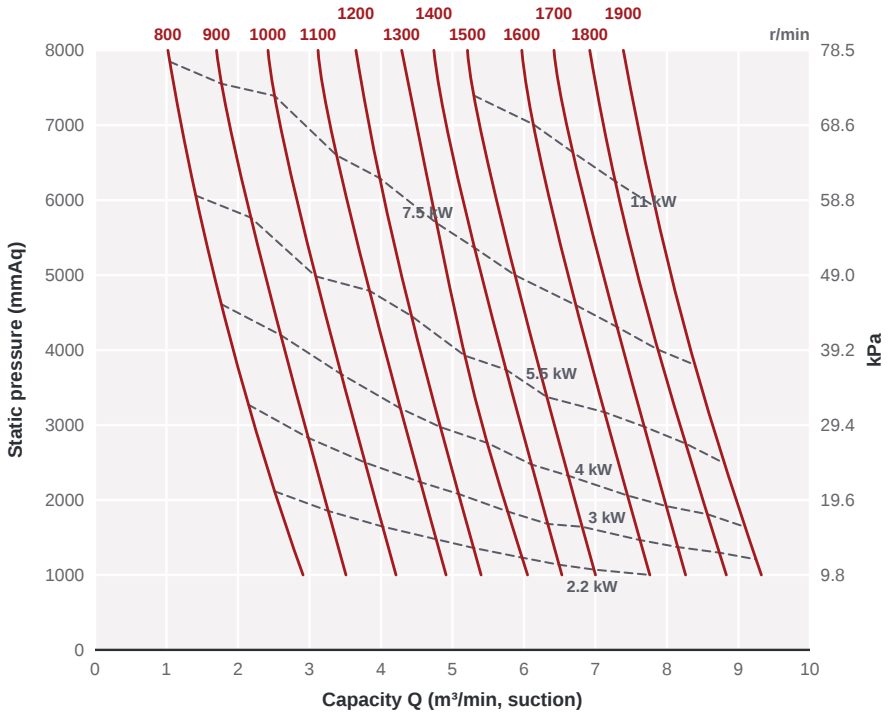
Bore 3" (80 mm) | 800 – 1900 r/min | pressure operation

PERFORMANCE TABLE

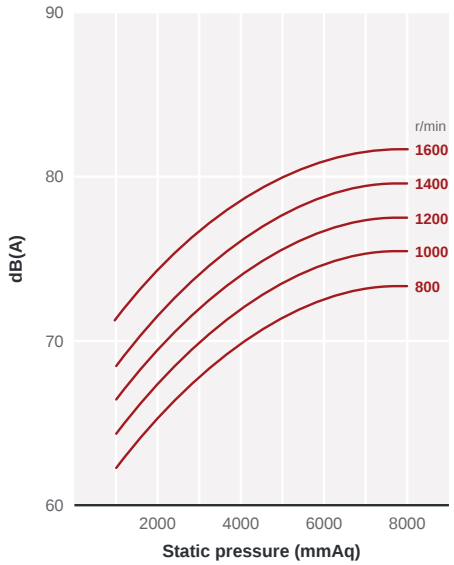
	1000 mmAq 9.8 kPa		2000 mmAq 19.6 kPa		3000 mmAq 29.4 kPa		4000 mmAq 39.2 kPa		5000 mmAq 49.0 kPa		6000 mmAq 58.8 kPa		7000 mmAq 68.6 kPa		8000 mmAq 78.5 kPa	
r/min	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW
800	2.91	1.40	2.55	2.12	2.23	2.81	1.93	3.51	1.67	4.31	1.42	5.43	1.21	6.55	1.02	7.67
900	3.51	1.50	3.25	2.32	2.93	3.14	2.63	3.84	2.36	4.64	2.12	5.76	1.89	6.88	1.70	8.00
1000	4.21	1.60	3.95	2.53	3.63	3.47	3.33	4.25	3.06	5.52	2.82	6.64	2.61	7.06	2.42	8.18
1100	4.91	1.70	4.65	2.75	4.33	3.80	4.03	4.70	3.76	5.71	3.52	6.83	3.31	7.95	3.12	9.07
1200	5.40	1.80	5.14	2.91	4.82	4.03	4.52	5.04	4.29	6.05	4.07	7.17	3.84	8.29	3.65	9.41
1300	6.05	1.90	5.75	3.20	5.43	4.25	5.13	5.60	4.95	6.72	4.69	7.84	4.48	8.96	4.29	10.08
1400	6.53	2.00	6.27	3.47	5.95	4.59	5.65	5.82	5.38	7.05	5.14	8.28	4.93	9.56	4.74	10.81
1500	7.00	2.10	6.74	3.50	6.42	5.04	6.12	6.27	5.85	7.50	5.61	8.96	5.40	10.42	5.21	11.88
1600	7.76	2.20	7.50	3.92	7.18	5.26	6.86	6.66	6.61	8.06	6.37	9.52	6.16	10.98	5.97	12.44
1700	8.26	2.32	8.00	4.15	7.67	5.53	7.36	7.04	7.08	8.53	6.83	10.04	6.61	11.54	6.42	13.04
1800	8.83	2.42	8.52	4.37	8.17	5.87	7.86	7.49	7.59	8.98	7.33	10.59	7.12	12.21	6.92	13.82
1900	9.32	2.52	8.97	4.83	8.64	6.20	8.33	7.82	8.06	9.43	7.81	11.15	7.59	12.63	7.39	14.14

Q: air volume at suction (m³/min). kW: required power including V-belt transmission — no extra margin needed when selecting the motor.

PERFORMANCE CURVES



NOISE LEVEL · dB(A) at 1 m



Dashed lines: motor power steps, interpolated from the power values of the table.

RL SERIES

RL-100

Three-lobe roots blower



KEY DATA

Bore	**4" (100 mm)**
Speed	**800 – 1900 r/min**
Capacity	**2.2 – 14.65 m³/min**
Pressure	**1000 – 8000 mmAq (9.8 – 78.5 kPa)**
Required power	**1.37 – 21.23 kW**
Noise	**66 – 84 dB(A) at 1 m**
Weight	**166 kg body / 330 kg complete set**

MODEL CODE

RL-100
Blower body only

RLC-100
Complete set, pressure operation

RLV-100
Complete set, vacuum operation

PERFORMANCE DATA

Bore 4" (100 mm) | 800 – 1900 r/min | pressure operation

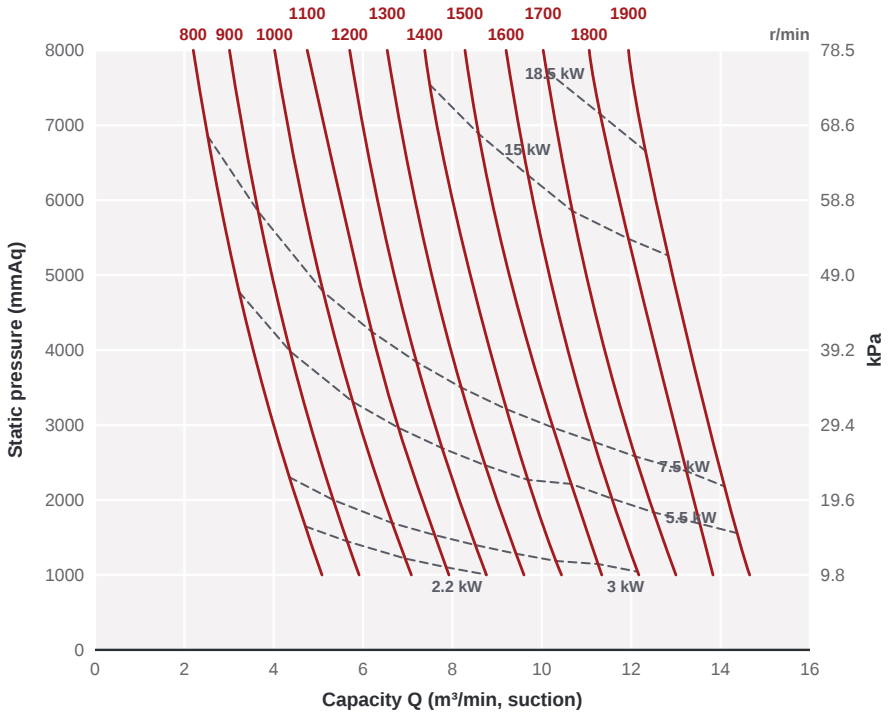
Redpump
Italy Standard

PERFORMANCE TABLE

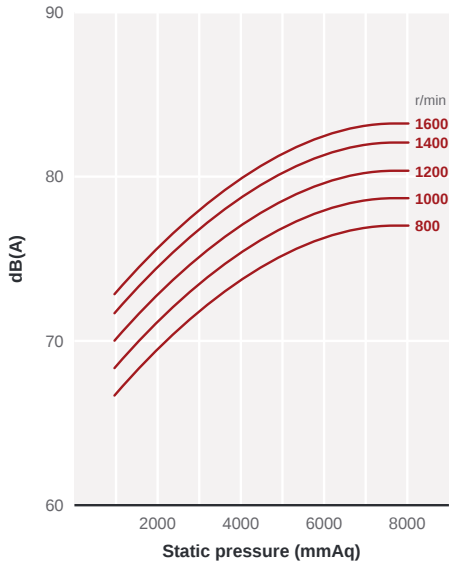
	1000 mmAq 9.8 kPa		2000 mmAq 19.6 kPa		3000 mmAq 29.4 kPa		4000 mmAq 39.2 kPa		5000 mmAq 49.0 kPa		6000 mmAq 58.8 kPa		7000 mmAq 68.6 kPa		8000 mmAq 78.5 kPa	
r/min	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW
800	5.08	1.37	4.51	2.65	4.00	3.80	3.54	4.88	3.14	5.68	2.79	6.78	2.48	7.62	2.20	8.43
900	5.91	1.55	5.33	3.00	4.82	4.31	4.36	5.52	3.96	6.55	3.60	7.67	3.29	8.63	3.01	9.54
1000	7.08	1.83	6.47	3.52	5.93	5.06	5.44	6.48	5.02	7.79	4.64	9.00	4.31	10.13	4.02	11.18
1100	7.92	2.01	7.31	3.87	6.77	5.56	6.28	7.12	5.86	8.67	5.48	9.90	5.15	11.14	4.75	12.29
1200	8.76	2.19	8.15	4.22	7.61	6.07	7.12	7.77	6.70	9.55	6.32	10.80	5.99	12.15	5.70	13.41
1300	9.60	2.38	8.99	4.57	8.45	6.58	7.96	8.42	7.54	10.43	7.16	11.70	6.83	13.17	6.54	14.53
1400	10.44	2.56	9.83	4.92	9.29	7.08	8.86	9.07	8.38	11.31	8.00	12.60	7.67	14.18	7.38	15.69
1500	11.34	2.67	10.75	4.93	10.23	7.59	9.75	9.72	9.25	12.22	8.88	13.50	8.56	15.19	8.28	16.75
1600	12.17	2.89	11.58	5.45	11.05	8.09	10.58	10.37	10.17	12.73	9.80	14.40	9.48	16.20	9.20	17.87
1700	13.00	3.12	12.41	5.97	11.88	8.60	11.40	11.01	11.00	13.23	10.63	15.30	10.31	17.21	10.03	18.99
1800	13.83	3.34	13.42	6.50	13.01	9.10	12.53	11.66	12.12	13.87	11.70	16.20	11.37	18.22	11.06	20.11
1900	14.65	3.57	14.15	7.03	13.75	9.61	13.32	12.31	12.91	14.25	12.54	17.10	12.22	19.23	11.94	21.23

Q: air volume at suction (m³/min). kW: required power including V-belt transmission — no extra margin needed when selecting the motor.

PERFORMANCE CURVES



NOISE LEVEL · dB(A) at 1 m



Dashed lines: motor power steps, interpolated from the power values of the table.

RL SERIES

RL-125

Three-lobe roots blower



KEY DATA

Bore	**5" (125 mm)**
Speed	**700 – 1800 r/min**
Capacity	**6.19 – 28.74 m³/min**
Pressure	**1000 – 8000 mmAq (9.8 – 78.5 kPa)**
Required power	**3.31 – 43.2 kW**
Noise	**71 – 90 dB(A) at 1 m**
Weight	**296 kg body / 485 kg complete set**

MODEL CODE

RL-125
Blower body only

RLC-125
Complete set, pressure operation

RLV-125
Complete set, vacuum operation

PERFORMANCE DATA

Bore 5" (125 mm) | 700 – 1800 r/min | pressure operation

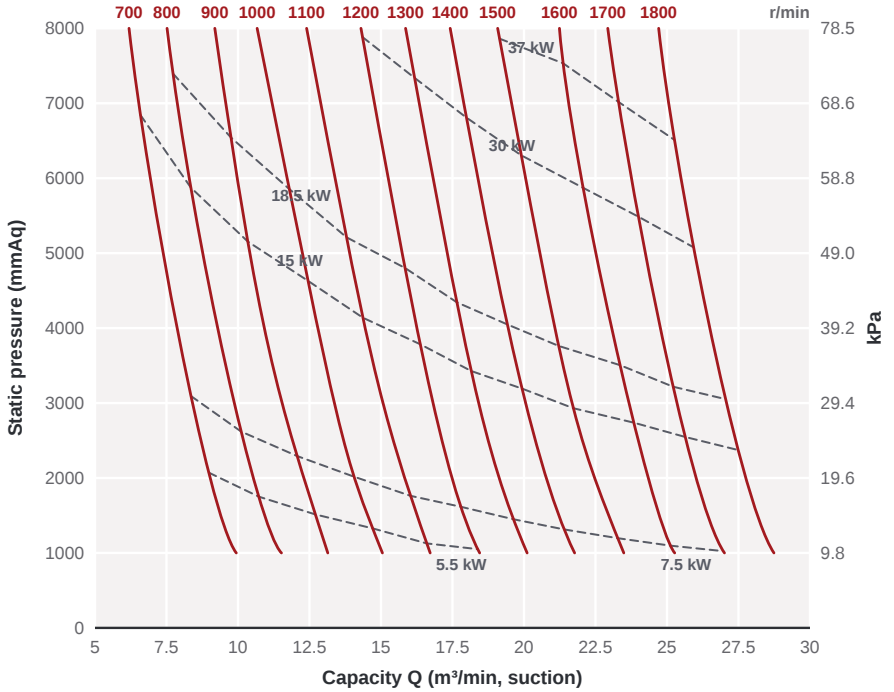
Redpump
Italy Standard

PERFORMANCE TABLE

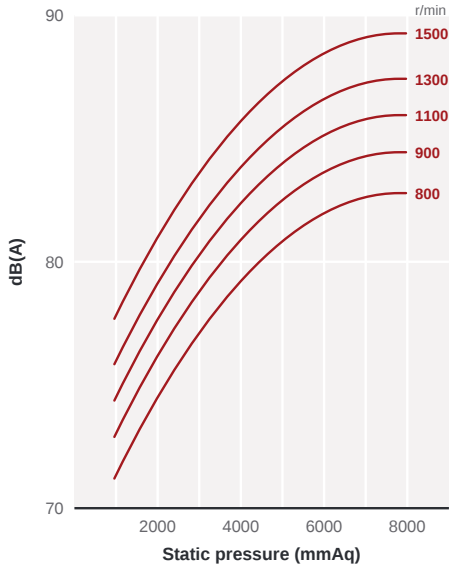
	1000 mmAq 9.8 kPa		2000 mmAq 19.6 kPa		3000 mmAq 29.4 kPa		4000 mmAq 39.2 kPa		5000 mmAq 49.0 kPa		6000 mmAq 58.8 kPa		7000 mmAq 68.6 kPa		8000 mmAq 78.5 kPa	
r/min	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW
700	9.94	3.31	9.06	5.36	8.38	7.31	7.90	9.35	7.40	11.31	6.91	13.35	6.55	15.31	6.19	17.27
800	11.52	3.78	10.57	6.07	9.84	8.38	9.32	10.68	8.79	12.98	8.27	15.29	7.86	17.59	7.52	19.89
900	13.14	4.20	12.29	6.73	11.56	9.36	10.93	11.98	10.40	14.61	9.99	17.13	9.56	19.65	9.19	22.18
1000	15.05	4.55	14.02	7.45	13.34	10.34	12.74	13.24	12.20	16.03	11.72	18.92	11.24	21.92	10.67	24.82
1100	16.72	5.10	15.80	8.25	15.10	11.29	14.42	14.55	13.90	17.81	13.38	21.07	12.92	24.98	12.40	28.90
1200	18.45	5.32	17.51	8.86	16.79	12.31	16.23	15.74	15.74	19.18	15.26	22.93	14.80	26.69	14.30	30.44
1300	20.11	5.77	19.15	9.62	18.40	13.27	17.87	17.32	17.32	20.75	16.79	24.81	16.37	28.88	15.86	32.19
1400	21.77	6.23	20.78	10.28	20.01	14.22	19.47	18.27	18.93	22.97	18.38	27.35	17.94	30.63	17.42	33.91
1500	23.49	6.65	22.50	10.97	21.73	15.30	21.05	19.50	20.51	24.38	20.06	28.82	19.61	32.69	19.08	37.69
1600	25.27	7.06	24.33	11.59	23.70	16.21	23.06	20.74	22.54	26.33	21.98	30.54	21.59	34.75	21.24	38.97
1700	27.01	7.38	26.05	12.35	25.32	17.19	24.79	23.21	24.27	27.43	23.73	32.71	23.31	36.92	22.93	41.43
1800	28.74	8.17	27.79	13.11	27.05	18.18	26.52	24.30	26.00	29.59	25.46	34.87	25.08	39.04	24.71	43.20

Q: air volume at suction (m³/min). kW: required power including V-belt transmission — no extra margin needed when selecting the motor.

PERFORMANCE CURVES



NOISE LEVEL · dB(A) at 1 m



Dashed lines: motor power steps, interpolated from the power values of the table.

RL SERIES

RL-150

Three-lobe roots blower



KEY DATA

Bore	**6" (150 mm)**
Speed	**700 – 1800 r/min**
Capacity	**8.66 – 37.26 m³/min**
Pressure	**1000 – 8000 mmAq (9.8 – 78.5 kPa)**
Required power	**3.99 – 56.57 kW**
Noise	**73 – 90 dB(A) at 1 m**
Weight	**352 kg body / 525 kg complete set**

MODEL CODE

RL-150
Blower body only

RLC-150
Complete set, pressure operation

RLV-150
Complete set, vacuum operation

PERFORMANCE DATA

Bore 6" (150 mm) | 700 – 1800 r/min | pressure operation

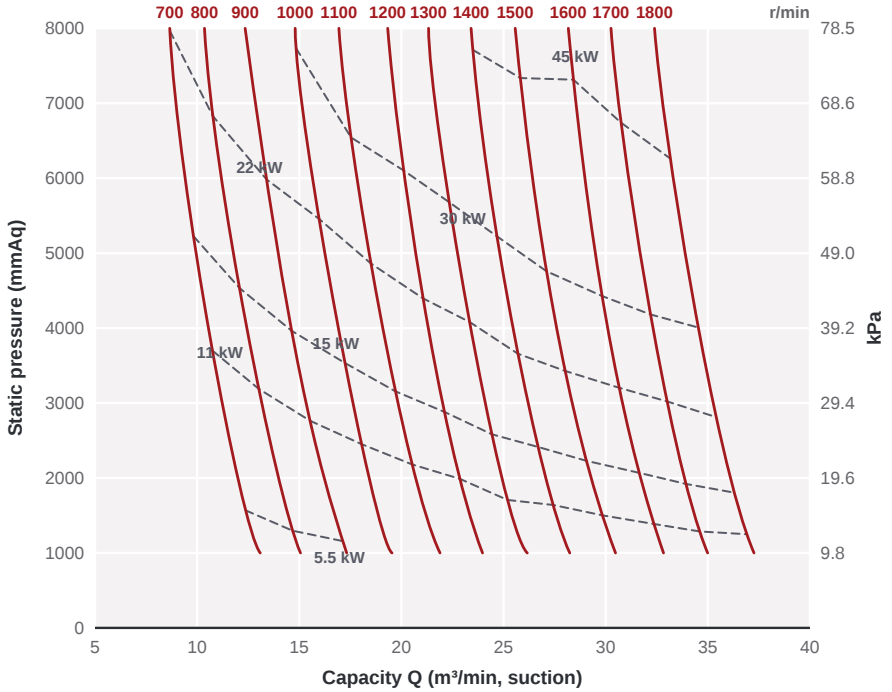
Redpump
Italy Standard

PERFORMANCE TABLE

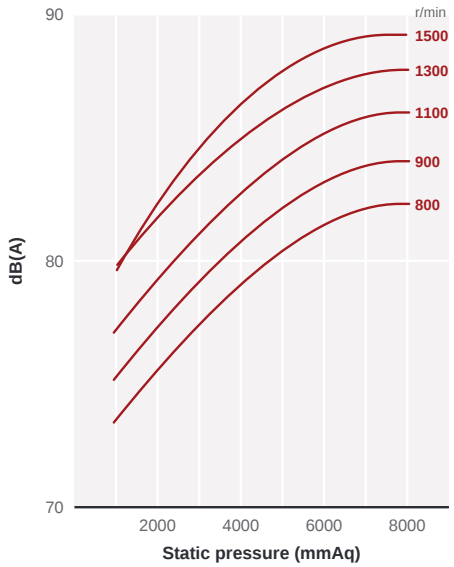
	1000 mmAq 9.8 kPa		2000 mmAq 19.6 kPa		3000 mmAq 29.4 kPa		4000 mmAq 39.2 kPa		5000 mmAq 49.0 kPa		6000 mmAq 58.8 kPa		7000 mmAq 68.6 kPa		8000 mmAq 78.5 kPa	
r/min	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW
700	13.09	3.99	11.97	6.65	11.25	9.21	10.54	11.77	9.93	14.42	9.41	16.98	8.98	19.54	8.66	22.09
800	15.06	4.62	13.96	7.59	13.19	10.44	12.41	13.41	11.76	16.38	11.21	19.45	10.71	22.53	10.36	25.60
900	17.32	4.96	16.22	8.38	15.34	11.81	14.56	15.12	14.01	18.43	13.35	22.06	12.89	25.70	12.35	29.34
1000	19.54	5.65	18.46	9.33	17.59	13.03	16.93	16.72	16.29	20.41	15.64	23.89	15.09	27.36	14.80	30.95
1100	21.89	6.15	20.75	10.26	19.84	14.36	19.04	18.47	18.47	22.57	17.78	27.38	17.31	32.20	16.94	37.00
1200	23.97	6.67	22.88	11.05	21.99	15.53	21.34	19.92	20.69	25.17	20.14	29.55	19.69	33.93	19.33	38.30
1300	26.16	7.19	24.93	12.58	24.14	16.73	23.35	21.56	22.79	26.95	22.12	31.44	21.67	35.93	21.33	40.42
1400	28.24	7.69	27.09	12.86	26.18	18.03	25.49	24.07	24.80	28.71	24.22	34.45	23.79	40.19	23.41	46.94
1500	30.48	8.26	29.32	13.72	28.39	19.31	27.57	25.59	26.99	31.41	26.41	37.22	25.93	43.04	25.57	48.86
1600	32.83	8.74	31.73	14.59	30.85	20.56	30.07	27.64	29.52	33.16	28.96	37.73	28.52	43.26	28.17	48.79
1700	34.99	9.19	33.88	15.50	33.00	21.85	32.34	28.79	31.67	35.44	31.11	40.97	30.61	46.41	30.26	52.05
1800	37.26	9.20	36.16	16.41	35.27	23.29	34.49	29.95	33.94	36.60	33.26	43.26	32.75	49.91	32.39	56.57

Q: air volume at suction (m³/min). kW: required power including V-belt transmission — no extra margin needed when selecting the motor.

PERFORMANCE CURVES



NOISE LEVEL · dB(A) at 1 m



Dashed lines: motor power steps, interpolated from the power values of the table.

RL-200

Three-lobe roots blower



KEY DATA

Bore	**8" (200 mm)**
Speed	**600 – 1400 r/min**
Capacity	**16.16 – 59.93 m³/min**
Pressure	**1000 – 8000 mmAq (9.8 – 78.5 kPa)**
Required power	**6.77 – 92.66 kW**
Noise	**76 – 94 dB(A) at 1 m**
Weight	**880 kg body / 1155 kg complete set**

MODEL CODE

RL-200
Blower body only

RLC-200
Complete set, pressure operation

RLV-200
Complete set, vacuum operation

PERFORMANCE DATA

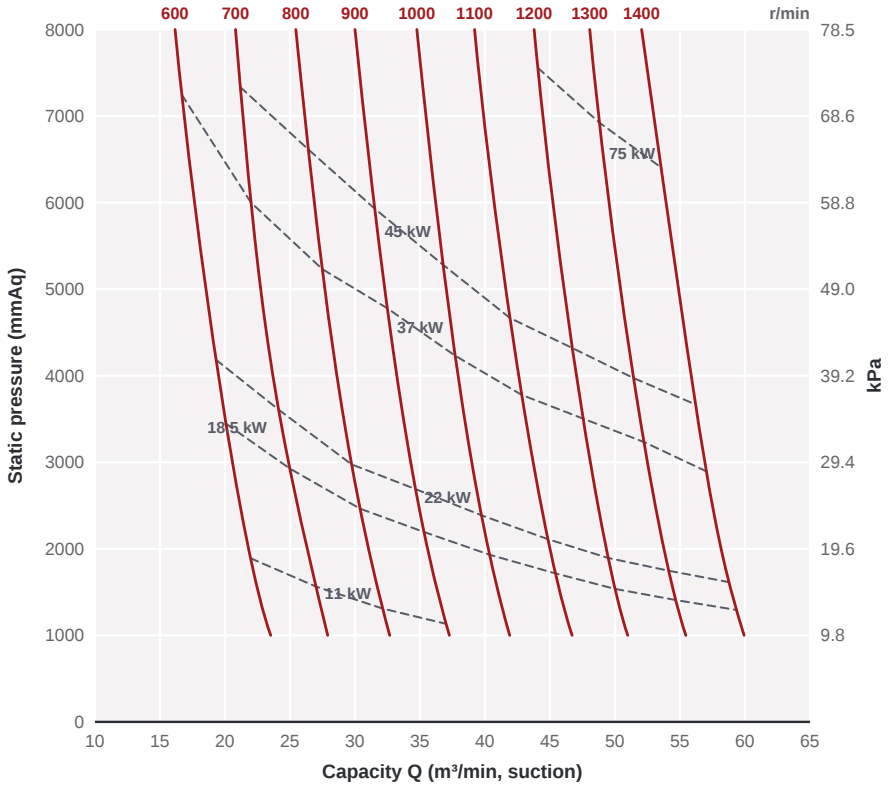
Bore 8" (200 mm) | 600 – 1400 r/min | pressure operation

PERFORMANCE TABLE

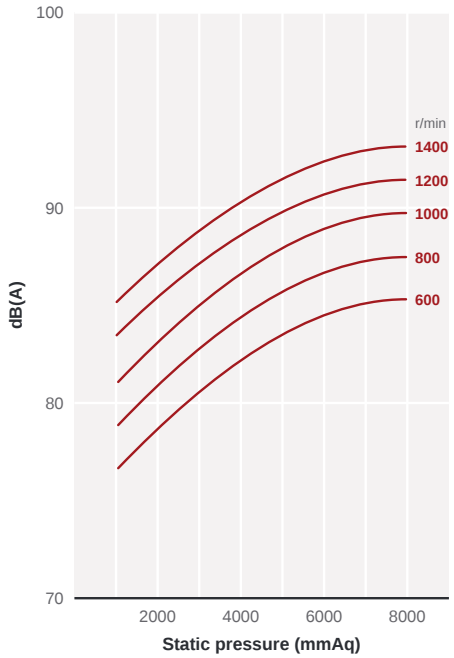
	1000 mmAq 9.8 kPa		2000 mmAq 19.6 kPa		3000 mmAq 29.4 kPa		4000 mmAq 39.2 kPa		5000 mmAq 49.0 kPa		6000 mmAq 58.8 kPa		7000 mmAq 68.6 kPa		8000 mmAq 78.5 kPa	
r/min	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW
600	23.52	6.77	21.87	11.52	20.52	16.26	19.45	21.29	18.58	25.16	17.61	30.97	16.84	35.81	16.16	40.65
700	27.90	7.90	26.20	13.40	24.90	18.90	23.80	24.00	22.80	30.00	21.90	37.00	21.48	44.00	20.81	47.00
800	32.67	9.06	31.04	15.33	29.69	22.17	28.63	27.95	27.66	35.66	26.89	41.45	26.12	47.23	25.45	53.01
900	37.26	10.06	35.61	17.13	34.26	24.19	33.19	30.97	32.32	38.71	31.45	45.48	30.68	52.26	30.00	59.03
1000	41.90	11.05	40.29	19.05	38.95	26.67	37.90	35.24	37.05	42.86	36.19	50.48	35.43	58.10	34.76	65.71
1100	46.70	12.20	45.00	21.00	43.70	30.00	42.60	39.00	41.60	48.00	40.70	56.00	39.90	65.00	39.20	74.00
1200	50.98	13.25	49.34	23.04	48.10	31.68	47.04	42.24	46.08	50.88	45.22	60.48	44.45	69.12	43.78	79.68
1300	55.45	14.28	53.77	24.62	52.49	34.47	51.41	45.30	50.42	55.15	49.54	65.00	48.75	75.83	48.06	86.67
1400	59.93	15.31	58.22	26.19	56.91	38.27	55.80	48.35	54.79	59.42	53.99	70.50	53.08	81.58	52.05	92.66

Q: air volume at suction (m³/min). kW: required power including V-belt transmission — no extra margin needed when selecting the motor.

PERFORMANCE CURVES



NOISE LEVEL · dB(A) at 1 m



Dashed lines: motor power steps, interpolated from the power values of the table.

RL SERIES

RL-250

Three-lobe roots blower



KEY DATA

Bore	**10" (250 mm)**
Speed	**600 – 1400 r/min**
Capacity	**23.71 – 80.68 m³/min**
Pressure	**1000 – 8000 mmAq (9.8 – 78.5 kPa)**
Required power	**8.82 – 124.19 kW**
Noise	**72 – 99 dB(A) at 1 m**
Weight	**1050 kg body / 1500 kg complete set**

MODEL CODE

RL-250
Blower body only

RLC-250
Complete set, pressure operation

RLV-250
Complete set, vacuum operation

PERFORMANCE DATA

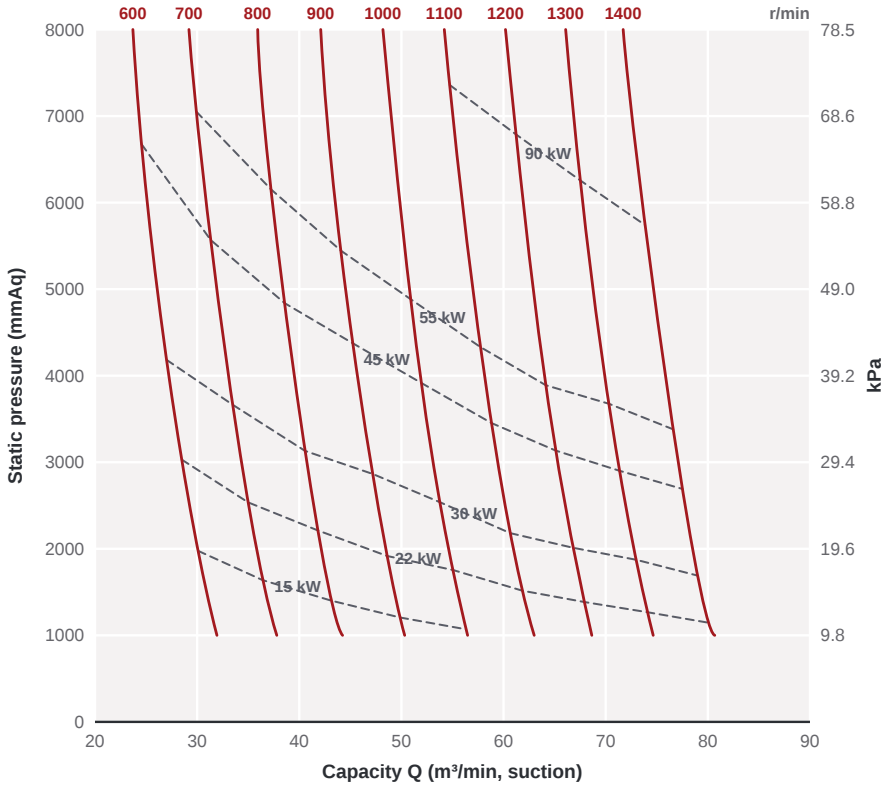
Bore 10" (250 mm) | 600 – 1400 r/min | pressure operation

PERFORMANCE TABLE

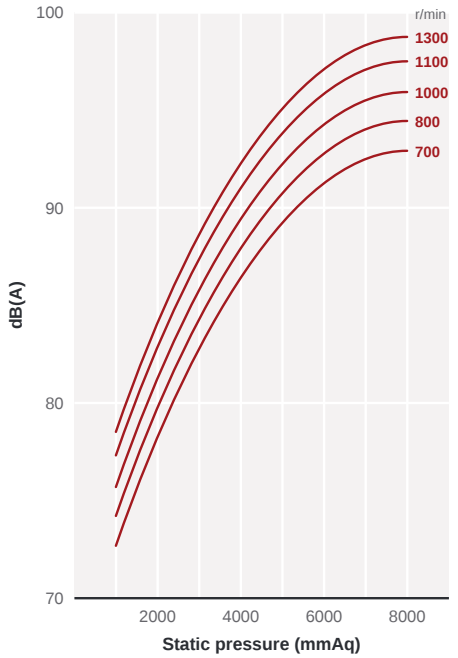
	1000 mmAq 9.8 kPa		2000 mmAq 19.6 kPa		3000 mmAq 29.4 kPa		4000 mmAq 39.2 kPa		5000 mmAq 49.0 kPa		6000 mmAq 58.8 kPa		7000 mmAq 68.6 kPa		8000 mmAq 78.5 kPa	
r/min	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW
600	31.94	8.82	30.00	15.15	28.55	21.78	27.29	29.05	26.13	34.24	25.06	41.50	24.37	46.68	23.71	51.87
700	37.80	10.29	35.90	17.69	34.30	25.73	33.00	32.16	32.00	40.74	30.90	48.24	29.90	54.67	29.20	61.10
800	44.24	11.57	42.22	20.14	40.67	28.92	39.52	37.20	38.46	46.50	37.40	53.73	36.43	62.00	35.93	70.27
900	50.32	12.97	48.39	22.82	46.94	31.12	45.77	41.50	44.52	50.84	43.45	60.17	42.77	69.50	42.10	78.83
1000	56.48	14.29	54.67	24.50	53.05	34.71	51.90	45.94	50.76	56.19	49.81	66.55	48.95	77.59	48.19	88.63
1100	63.00	15.75	61.00	27.87	59.40	39.66	58.10	51.46	57.00	62.18	56.00	73.97	55.00	85.76	54.20	97.55
1200	68.64	17.09	66.82	29.85	65.28	43.22	64.03	56.46	62.88	66.89	61.92	80.28	61.06	92.62	60.19	104.93
1300	74.65	18.48	72.78	31.67	71.20	46.46	69.92	59.09	68.74	72.85	67.76	86.57	66.87	99.94	66.08	113.36
1400	80.68	19.86	78.56	34.55	77.05	49.66	75.90	63.71	74.63	78.82	73.63	93.93	72.52	109.08	71.71	124.19

Q: air volume at suction (m³/min). kW: required power including V-belt transmission — no extra margin needed when selecting the motor.

PERFORMANCE CURVES



NOISE LEVEL · dB(A) at 1 m



Dashed lines: motor power steps, interpolated from the power values of the table.

RL SERIES

RL-300

Three-lobe roots blower



KEY DATA

Bore	**12" (300 mm)**
Speed	**600 – 1400 r/min**
Capacity	**55.3 – 178 m³/min**
Pressure	**1000 – 8000 mmAq (9.8 – 78.5 kPa)**
Required power	**18.1 – 250.5 kW**
Noise	**90 – 111 dB(A) at 1 m**
Weight	**2000 kg body / 2800 kg complete set**

MODEL CODE

RL-300
Blower body only

RLC-300
Complete set, pressure operation

RLV-300
Complete set, vacuum operation

PERFORMANCE DATA

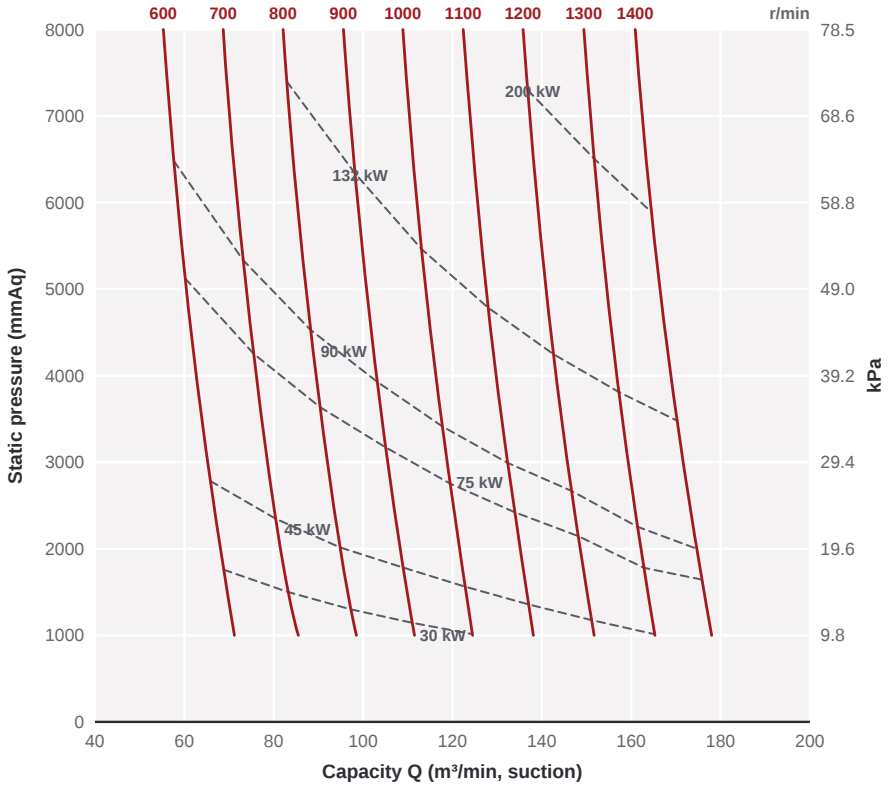
Bore 12" (300 mm) | 600 – 1400 r/min | pressure operation

PERFORMANCE TABLE

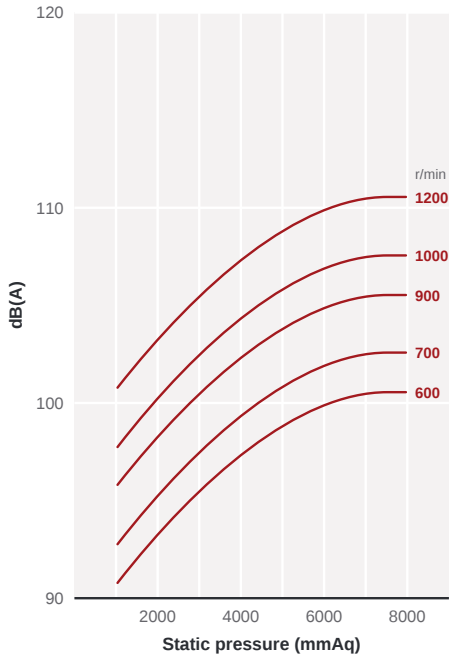
	1000 mmAq 9.8 kPa		2000 mmAq 19.6 kPa		3000 mmAq 29.4 kPa		4000 mmAq 39.2 kPa		5000 mmAq 49.0 kPa		6000 mmAq 58.8 kPa		7000 mmAq 68.6 kPa		8000 mmAq 78.5 kPa	
r/min	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW	Q	kW
600	71.2	18.1	68.0	33.8	65.3	48.1	62.6	61.4	60.5	73.6	58.5	84.9	56.8	95.5	55.3	105.2
700	85.5	20.8	81.4	39.2	78.7	55.9	76.0	71.4	73.9	85.6	71.9	98.8	70.2	111.1	68.7	122.5
800	98.5	23.7	94.8	44.7	92.1	63.8	89.4	81.4	87.3	97.7	85.3	112.7	83.6	126.8	82.1	139.8
900	111.5	26.6	108.2	50.1	105.6	71.6	102.9	91.4	100.8	109.7	98.8	126.7	97.1	142.4	95.6	157.1
1000	124.5	29.6	121.6	57.0	118.9	80.9	116.2	102.9	114.1	123.3	112.1	142.1	110.4	159.6	108.9	176.0
1100	138.1	34.5	135.1	63.9	132.4	90.2	129.7	114.4	127.6	136.8	125.6	157.6	123.9	176.9	122.4	194.8
1200	151.7	39.5	148.5	70.8	145.8	99.5	143.1	125.9	141.0	150.4	139.0	171.9	137.3	194.1	135.8	213.7
1300	165.3	44.5	162.1	83.6	159.4	108.8	156.7	137.4	154.6	164.0	152.6	188.4	150.9	211.3	149.4	232.6
1400	178.0	47.9	174.6	90.0	171.7	117.2	168.8	148.0	166.5	176.6	164.3	202.9	162.5	227.6	160.9	250.5

Q: air volume at suction (m³/min). kW: required power including V-belt transmission — no extra margin needed when selecting the motor.

PERFORMANCE CURVES



NOISE LEVEL · dB(A) at 1 m



Dashed lines: motor power steps, interpolated from the power values of the table.

Selection guide

NOISE LEVEL

Measuring method: the noise values are the average values measured at a distance of 1 m from the blower under standard conditions.

Noise depends on blower size, discharge pressure and speed. To reduce it further: fit silencers on the inlet and outlet; lower the blower speed; use expansion joints or rubber vibration isolators to prevent surging and vibration.

UNITS

1 kPa = 1000 Pa = 1000 N/m²

1 bar = 1000 mbar = 100 kPa

1 mbar = 100 N/m²

HOW TO USE THE PERFORMANCE TABLES

The performance tables give the model, bore, speed (r/min), static pressure, air volume and required power of the motor.

1. The air volume in the performance tables is the volume of intake air at standard conditions: temperature 20 °C, absolute pressure 1 atm (10333 mmAq) and relative humidity 65 %.
2. Air volume is generally given in one of three ways, which can be converted into each other: (1) at suction condition — at the suction pressure, temperature and humidity; (2) at standard condition; (3) at base condition — 0 °C, absolute pressure 1 atm, dry air.
3. Conversion: Q2 = Q1 × P1 / P2 × T2 / T1, where Q1 is the air volume (m³/min) at absolute pressure P1 (mmAq) and absolute temperature T1 (K), and Q2 the air volume at P2 and T2.
4. With the air volume and static pressure found above, select the model, bore, speed and required power from the performance table.
5. Several models may cover the same duty. For economy choose the smaller blower; for a lower sound level choose the larger blower running at lower speed.
6. The power values in the tables are the required power including the V-belt transmission, so no extra margin is needed when selecting the motor.
7. Motor output: kW = √3 × V × A × cos φ × Eff ÷ 1000 (V: voltage, A: current, cos φ: power factor, Eff: motor efficiency).

PRESSURE CONVERSION

Pressure	atm	kPa	mbar	psi (lbf/in²)	kgf/cm²	in Hg	ft Aq	mmHg(Torr)	mmAq
1 atm	1	101.325	1013.25	14.696	1.0333	29.921	33.914	760	10333
1 kPa	0.0099	1	10	0.145	0.0102	0.295	0.335	7.5	102
1 mbar	0.00099	0.1	1	0.0145	0.00102	0.0295	0.0335	0.75	10.198
1 psi	0.0680	6.894	68.965	1	0.0703	2.036	2.308	51.71	703
1 kgf/cm²	0.968	98.062	980.392	14.228	1	28.96	32.82	735.53	10000
1 in Hg	0.0334	3.3863	33.898	0.491	0.0345	1	1.133	25.4	345.3
1 ft Aq	0.0295	2.99	29.851	0.434	0.0305	0.882	1	22.42	304.8
1 mmHg(Torr)	0.00132	0.1333	1.333	0.0193	0.00136	0.0394	0.0446	1	13.6
1 mmAq	0.000097	0.0098	0.09803	0.00142	0.0001	0.00290	0.00328	0.0736	1

CAPACITY CONVERSION

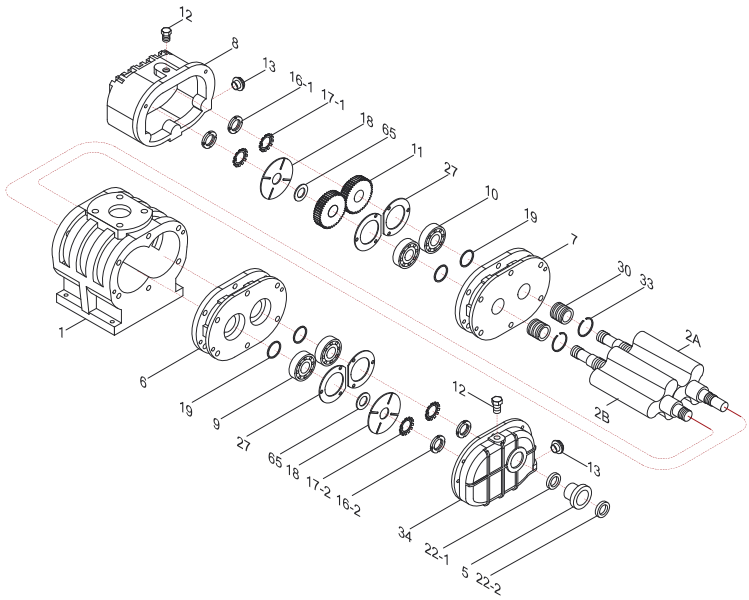
Capacity	m³/min	l/min	cm³/sec	in³/sec	CFM	U.S. GPM
1 m³/min	1	1000	16667	1016	35.288	264.172
1 l/min	0.001	1	16.67	1.02	0.0353	0.2641
1 cm³/sec	0.00006	0.06	1	0.061	0.00212	0.0158
1 in³/sec	0.00098	0.983	16.39	1	0.0347	0.2589
1 CFM	0.0283	28.32	471.957	28.8	1	7.4836
1 U.S. GPM	0.003785	3.7854	63.0915	3.8460	0.13360	1

Structure

MATERIALS

FC25	grey cast iron
SCM440	Cr-Mo alloy steel
SNCM220	Ni-Cr-Mo gear steel
SUJ2	bearing steel
S45C	carbon steel
SS41	structural steel
VITON	fluoro-rubber

RL STRUCTURAL DRAWING & PARTS LIST



PARTS LIST

No.	Name	Material	Q'ty	Fitting model
1	Casing	FC25	1	–
2A	Rotor + Drive Shaft	FC25 + SCM440	1	–
2B	Rotor + Driven Shaft	FC25 + SCM440	1	–
5	Seal Cover	FC25	4	RL-80 ~ 300
6	Front Side Cover	FC25	1	–
7	Rear Side Cover	FC25	1	–
8	Gear Box	FC25	1	–
9	Front Bearing	SUJ2	2	–
10	Rear Bearing	SUJ2	2	–
11	Timing Gear	SNCM220	2	–
12	Lubrication Plug	S45C	2	–
13	Oil Gauge	GLASS	2	–
16-1	Lock Nut	S45C	2	–
16-2	Lock Nut	S45C	2	–
17-1	Spring Washer	SS41	2	–
17-2	Spring Washer	SS41	2	–
18	Oil Splash	SS41	2	–
19	V-seal	VITON	4	–
22-1	Oil Seal	VITON	1	RL-40 ~ 300
22-2	Oil Seal	VITON	1	RL-125 ~ 300
27	Bearing Cover	SS41	4	–
30	Bearing Holder	SCM440	2	RL-150, RL-300
33	Piston Ring	FC25	2	Optional
34	Front Cover	FC25	1	–
65	Washer	SS41	2	–

Dimensions

NOTES

Unit: mm.

Wt: weight of blower body (kg).

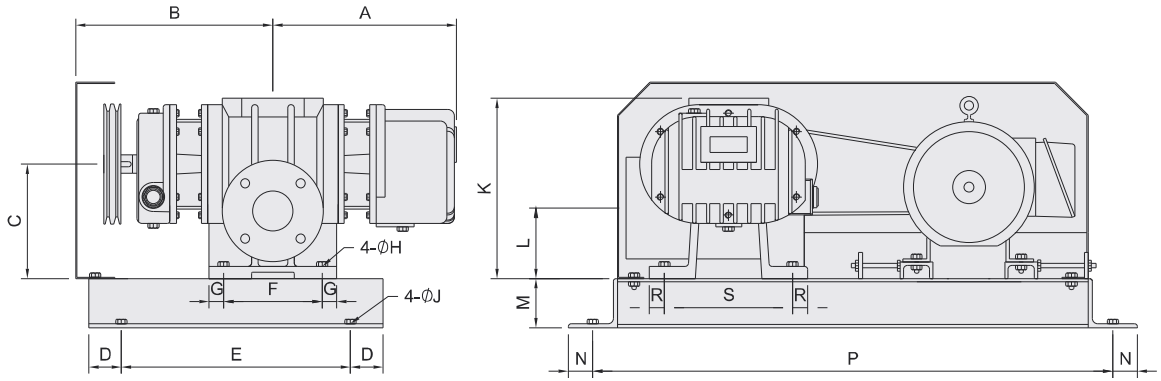
RLC: weight of the complete set for pressure operation (kg).

4-ØH: foot holes of the blower.

4-ØJ: anchor holes of the base plate.

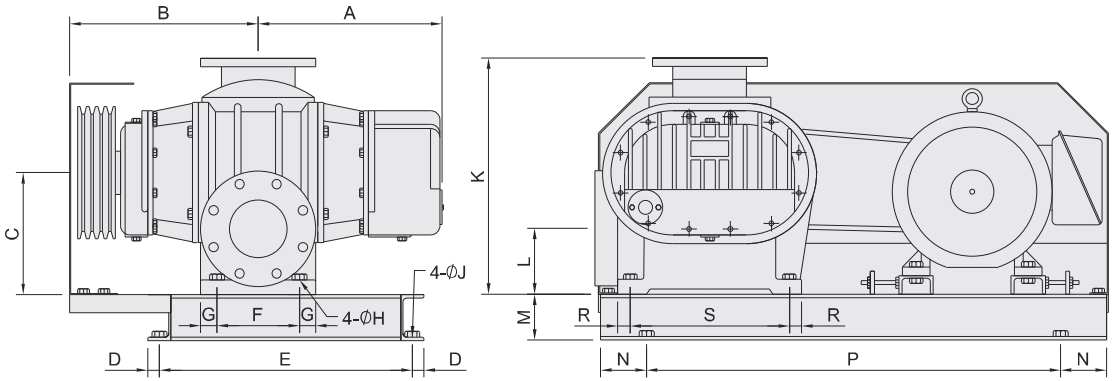
OUTLINE DIMENSIONS

RL-40, RL-50, RL-65



Type	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	S	Wt kg	RLC kg
RL-40	238	260	180	20	340	80	15	14	14	290	104	46	41.5	517	23	196	65	103
RL-50	278	302	175	50	350	150	22.5	14	14	276	103.5	75	37.5	795	22	196	70	136
RL-65	308	323	173	50	350	195	30.5	14	14	276	104	75	37.5	795	25	200	75	155

RL-80, RL-100, RL-125, RL-150



Type	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	S	Wt kg	RLC kg
RL-80	320	406	213	50	450	195	22.5	16	14	341	106.5	75	37.5	945	20	200	143	250
RL-100	320	467	225	25	550	282	20.5	16	19	354	114.5	100	100	760	30	232	166	330
RL-125	455	500	265	25	550	310	30	19	19	498	157.5	100	100	900	22.5	355	296	485
RL-150	510	566	295	25	550	428	26	19	19	457	163	100	100	900	22.5	355	352	525

Dimensions

NOTES

Unit: mm.

Wt: weight of blower body (kg).

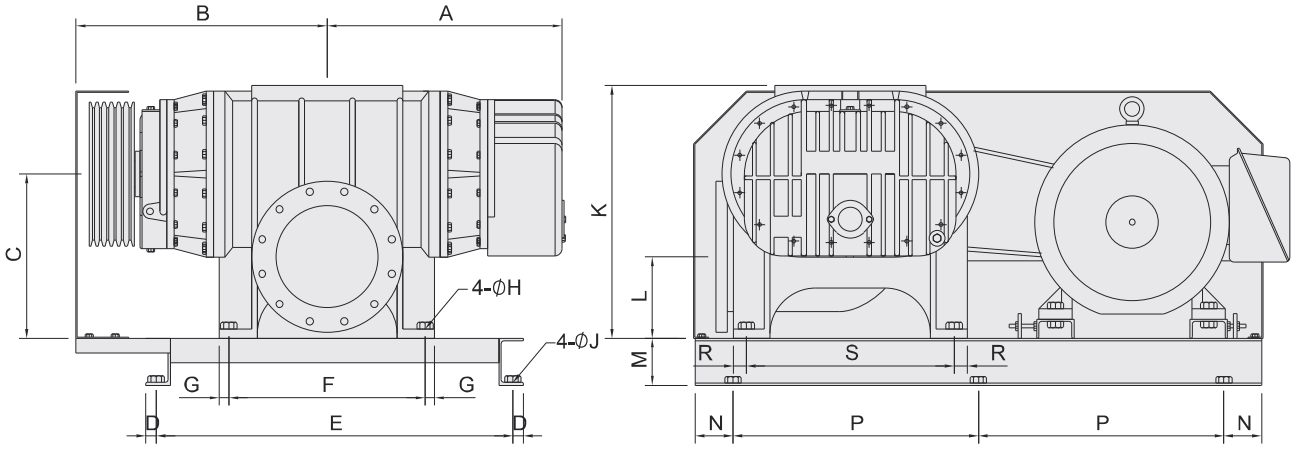
RLC: weight of the complete set for pressure operation (kg).

4-ØH: foot holes of the blower.

4-ØJ: anchor holes of the base plate.

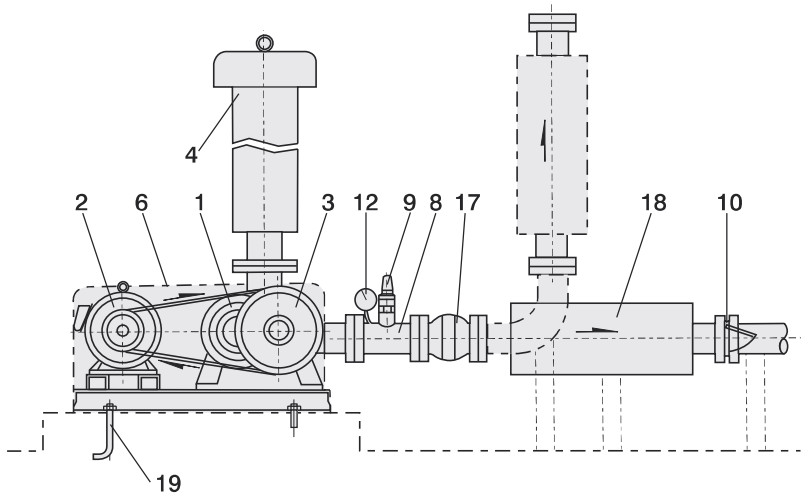
OUTLINE DIMENSIONS

RL-200, RL-250, RL-300



Type	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	S	Wt kg	RLC kg
RL-200	547	620	405	32.5	935	400	40	23	24	705	194	125	100	650	33	550	880	1155
RL-250	622	680	435	32.5	935	520	25	23	24	675	216	125	100	650	35	550	1050	1500
RL-300	685	828	550	35	1035	510	30	23	24	1010	300	250	235	1000	35	780	2000	2800

REFERENCE DRAWING FOR PIPING (EXAMPLE)



Note: install the check valve on horizontal piping.

No.	Accessories
1	Blower body
2	Motor
3	Blower pulley
4	Suction silencer set
6	Belt cover
8	T-type reducer
9	Pressure safety valve
10	Check valve
12	Pressure gauge
17	Expansion joint
18	Discharge silencer
19	Anchor bolt

Accessories

Unit: mm

ON THIS PAGE

Blower pulley (No. 3)
Pressure safety valve (No. 9)
Suction filter tank (No. 16)
Suction & discharge silencer set (No. 4)
Anchor bolt (No. 19)

NOTES

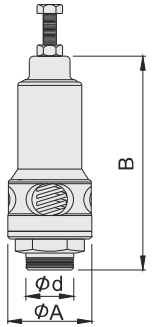
Unit: mm. Flanges JIS 10K. Fitting model shows the RL models each accessory fits.

DIMENSIONS OF ACCESSORIES

BLOWER PULLEY (No. 3)

TYPE	BORE	O.D.
SPB170 x 1G	40A	170mm
SPB180 x 2G	50A	180mm
SPB180 x 2G	65A	180mm
SPB200 x 3G	80A	200mm
SPB200 x 4G	100A	200mm
SPB280 x 4G	125A	280mm
SPB280 x 4G	150A	280mm
SPB375 x 5G	200A	375mm
SPB375 x 6G	250A	375mm
SPC450 x 8G	300A	450mm

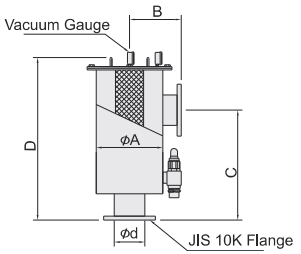
PRESSURE SAFETY VALVE (No. 9)



TYPE	ØA	B	Ød	Fitting model	Wt (kg)
PSV 32	71.5	182	PF 1 1/4"	RLC-40, 50, 65, 80	2
PSV 50	85	197	PF 2"	RLC-100, 125, 150	3
PSV 80	115	221	PF 3"	RLC-200, 250, 300	5.5

SUCTION FILTER TANK (No. 16)

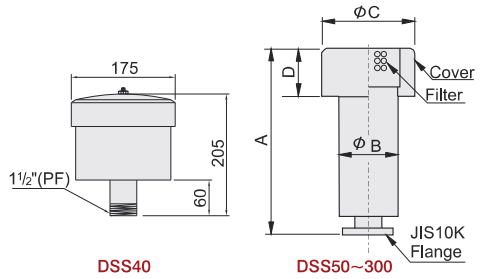
– vacuum gauge included



TYPE	BORE	A	B	C	D	Ød
SFT50	50A	244	222	340	530	60
SFT65	65A	244	222	440	630	76
SFT80	80A	244	222	525	730	89
SFT100	100A	285	240	525	730	114
SFT125	125A	360	280	505	730	139
SFT150	150A	360	280	595	880	165
SFT200	200A	470	335	650	1000	216
SFT250	250A	470	335	800	1200	267
SFT300	300A	600	405	1000	1400	319

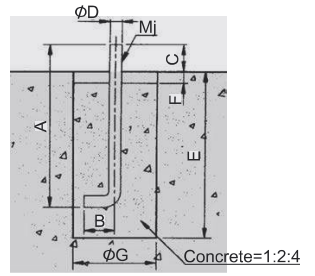
SUCTION & DISCHARGE SILENCER SET (No. 4)

– with air filter & cover



TYPE	BORE	A	B	C	D	Wt (kg)
DSS50	50A	548	140	220	80	7.5
DSS65	65A	548	165	235	85	14
DSS80	80A	818	210	310	145	22
DSS100	100A	818	240	350	160	26
DSS125	125A	1310	280	400	15	30
DSS150	150A	1310	300	450	190	43
DSS200	200A	1500	400	500	210	85
DSS250	250A	1700	400	500	210	110
DSS300	300A	1824	500	650	350	130

ANCHOR BOLT (No. 19)



TYPE	Mj	A	B	C	D	E	F	G	Fitting model
AB1	M12	165	30	25	12	165	5	100	RL-50, 65, 80
AB2	M16	245	45	40	16	245	5	120	RL-100, 125
AB3	M20	325	55	55	20	325	5	150	RL-150, 200, 250, 300

Accessories

Unit: mm

ON THIS PAGE

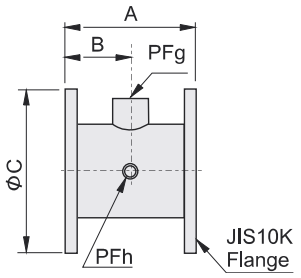
T-type reducer (No. 8)
Check valve (No. 10)
90° elbow (No. 14)
Discharge silencer (No. 18)
Expansion joint — EJA (rubber) (No. 17)

NOTES

Unit: mm. Flanges JIS 10K. Fitting model shows the RL models each accessory fits.

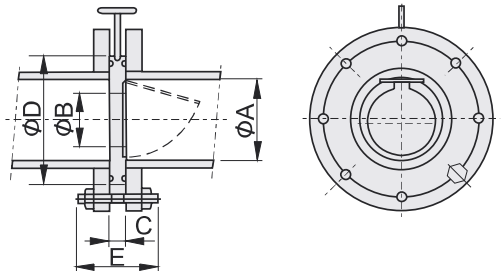
DIMENSIONS OF ACCESSORIES

T-TYPE REDUCER (No. 8)



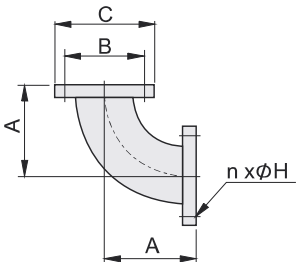
TYPE	BORE	A	B	C	PFg	PFh	Wt (kg)
TTR50	40A.50A	150	75	155	PF 1 1/2"	PF 1/4"	5
TTR65	65A	150	75	175	PF 1 1/4"	PF 1/4"	7
TTR80	80A	180	90	185	PF 1 1/2"	PF 1/4"	9
TTR100	100A	180	90	210	PF 1 1/2"	PF 1/4"	11
TTR125	125A	250	125	250	PF 1 1/2"	PF 1/4"	18
TTR150	150A	250	125	280	PF 1 1/2"	PF 1/4"	19
TTR200	200A	250	125	330	PF 3"	PF 1/4"	26
TTR250	250A	250	125	400	PF 3"	PF 1/4"	31
TTR300	300A	300	150	450	PF 3"	PF 1/4"	36

CHECK VALVE (No. 10)



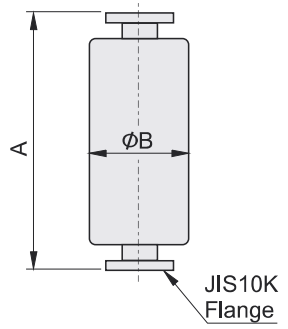
TYPE	BORE	A	B	C	D	E	Wt (kg)
NRV50	40A.50A	52.7	25	18	104	86	1
NRV65	65A	69.9	36	19	124	108	1.5
NRV80	80A	78.1	46	20	134	108	2
NRV100	100A	102.3	67	22	159	108	2.5
NRV125	125A	126.6	88	25	190	115	3.5
NRV150	150A	151.0	108	25	220	115	5
NRV200	200A	200.0	138	25	270	127	7
NRV250	250A	251.4	185	26	331	127	10
NRV300	300A	300.0	220	30	370	150	15

90° ELBOW (No. 14)



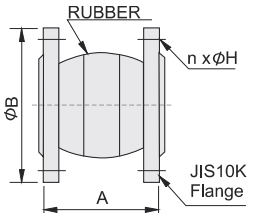
TYPE	BORE	A	B	C	n-ØH
NDE50	40A.50A	95.3	120	155	4-Ø19
NDE65	65A	95.3	140	175	4-Ø19
NDE80	80A	114.3	150	185	8-Ø19
NDE100	100A	152.4	175	210	8-Ø19
NDE125	125A	190.5	210	250	8-Ø19
NDE150	150A	228.6	240	280	8-Ø19
NDE200	200A	304.8	290	330	12-Ø19
NDE250	250A	381.0	355	400	12-Ø19
NDE300	300A	457.2	400	445	12-Ø19

DISCHARGE SILENCER (No. 18)



TYPE	BORE	A	B	Wt (kg)
DSF50	40A.50A	560	140	12
DSF65	65A	640	165	16
DSF80	80A	850	210	20
DSF100	100A	950	240	28
DSF125	125A	1150	280	33
DSF150	150A	1350	300	41
DSF200	200A	1600	400	90
DSF250	250A	1700	400	120
DSF300	300A	1800	500	140

EXPANSION JOINT — EJA (RUBBER) (No. 17)



TYPE	BORE	A	B	n-ØH	EJA Wt (kg)
EJA50	40A.50A	85	155	4-Ø19	4
EJA65	65A	95	175	4-Ø19	5
EJA80	80A	110	185	8-Ø19	6
EJA100	100A	115	210	8-Ø19	6
EJA125	125A	140	250	8-Ø19	10
EJA150	150A	150	280	8-Ø19	13
EJA200	200A	170	330	12-Ø19	16
EJA250	250A	210	400	12-Ø19	18
EJA300	300A	240	445	16-Ø19	21

Accessories

Unit: mm

ON THIS PAGE

Vacuum safety valve (No. 15)

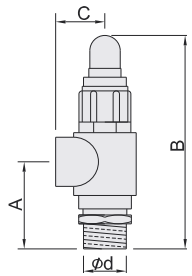
Expansion joint — EJB (SUS 304) (No. 17)

NOTES

Unit: mm. Flanges JIS 10K. Fitting model shows the RL models each accessory fits.

DIMENSIONS OF ACCESSORIES

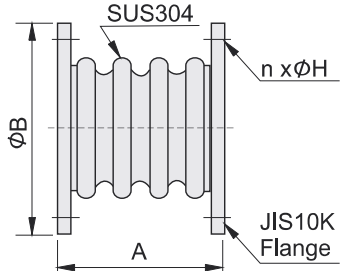
VACUUM SAFETY VALVE (No. 15)



TYPE	A	B	C	Ød	E	Wt (kg)	Fitting model
VSV 32	83	208	51	32	PT 1 1/4"	1.87	RLV-40, 50, 65, 80
VSV 50	103	266	67	50	PT 2"	3.91	RLV-100, 125, 150, 200, 250, 300

EXPANSION JOINT — EJB (SUS 304) (No. 17)

EJB type is recommended for pressure over 5000 mmAq.



TYPE	BORE	A	B	n-ØH	EJB Wt (kg)
EJB50	40A.50A	85	155	4-Ø19	4
EJB65	65A	95	175	4-Ø19	5
EJB80	80A	110	185	8-Ø19	6
EJB100	100A	115	210	8-Ø19	6
EJB125	125A	140	250	8-Ø19	10
EJB150	150A	150	280	8-Ø19	13
EJB200	200A	170	330	12-Ø19	16
EJB250	250A	210	400	12-Ø19	18
EJB300	300A	240	445	16-Ø19	21

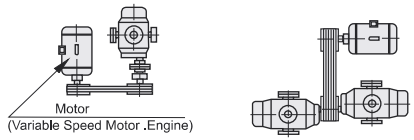
Drive & applications

DRIVING METHOD

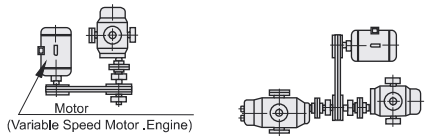
Motor with V-belt drive is standard. Direct coupling to the motor, engine drive with reduction gear, or variable-speed drive can also be supplied. For power above 90 kW (120 HP), V-belt drive with a counter shaft becomes standard.

DRIVING METHOD & APPLICATION GUIDE

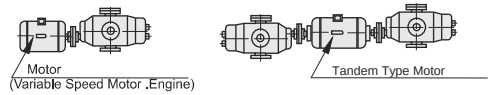
V-belt Driven



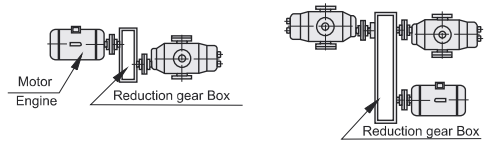
V-belt Counter Shaft Driven



Direct Coupling



Direct Coupling with Reduction Gear Box



APPLICATION GUIDE

The following applications are typical uses of Redpump roots blowers. This is a general listing and does not include all types of application.

PRESSURE OPERATION

Combustion air for burners, gas burners, coke-fired forges and small foundry cupolas in

- iron and steel industries
- food processing
- chemical plants
- non-ferrous metal refineries
- smelters and foundries
- bakeries
- glass, ceramics, brick
- oil, tar and asphalt heaters
- on-road machinery
- pipe wrapping machines

Aerating and agitating liquids in

- wastewater treatment plants
- water treatment
- electrotyping and engraving
- oyster washing
- oil blending
- ice manufacture
- chemical processes
- preventing ice formation in power plants
- fish ponds, lakes
- flotation
- blending vinegars
- asphalt refining
- food washing

Drying

- air and sand blasting
- grain conditioning in elevator storage
- supercharging
- dusting and spraying
- washing, drying and cleaning drums
- blending and mixing
- sand filter regeneration
- dust blowing
- cake blowing
- pressurizing paper mill headboxes

Pneumatic conveying

- granular materials
- insulation blowing
- liquid transfer
- fluidizing
- pulp conveying
- wood and metal chip conveying
- rock dusting
- flour and powder

VACUUM OPERATION

- industrial and commercial vacuum cleaning
- paper folding machinery
- garment presses
- bottle and tube filling machines
- drying
- mat formers
- pump priming
- dust collecting